Tingsong Xiao

xtstars1998@gmail.com | Homepage | Google Scholar | GitHub | LinkedIn

LLM Agents | **Post-training** | **World Models** — Full-time availability: 2027

Education

University of Florida Aug 2022 – Dec 2026
Ph.D. in Computer Science | GPA: 3.86/4.00

University of Electronic Science and Technology of China Aug 2017 – Jun 2021
B.Eng. in Computer Science | GPA: 3.92/4.00

Industry Experience

Amazon | Applied Scientist Intern May 2026 – Aug 2026

- Worked on the Amazon **Alexa AI** team, developing **LLM agentic planning** for **long-horizon tasks**, using **test-time search** and cross-trajectory feedback to guide multi-step decisions; explored **world models** for simulating action outcomes.
- Built **evaluation pipelines** to analyze task completion, constraint satisfaction, failure modes, and search efficiency.

eBay | Applied Researcher Intern May 2025 – Aug 2025

- Developed compact LLMs for product attribute extraction with $>$**90% F1 across 10 categories** and $>$**5**$\times$ **inference speedup**, using **supervised fine-tuning** and **knowledge distillation**.
- Constructed **domain-specific training datasets** from product descriptions using LLM-generated aspect–value labels; evaluated extraction quality and robustness on rare attributes.

Open-Source Contribution

Qwen-AgentWorld | **Stars: 1k+**
An open-source **language world model** that simulates interactive environments for training and evaluating **general-purpose AI agents**.

Selected Doctoral Research

University of Florida | Graduate Research Assistant Aug 2022 – Present

- Developed TD-HNODE (ICLR 2026) for continuous-time disease progression and XTSFormer (AAAI 2025) for multi-scale event prediction; XTSFormer achieved 5$\times$ faster training and 3.3 percentage-point accuracy/F1 gains.

Selected Publications

Peer-reviewed

- Temporally Detailed Hypergraph Neural ODE for Disease Progression Modeling. **ICLR 2026**.
- DecoyDB: A Dataset for Graph Contrastive Learning in Protein-Ligand Binding Affinity Prediction. **NeurIPS 2025**.
- CoastalBench: A Decade-Long High-Resolution Dataset to Emulate Complex Coastal Processes. **ICML 2025**.
- XTSFormer: Cross-Temporal-Scale Transformer for Irregular-Time Event Prediction in Clinical Applications. **AAAI 2025**.
- Spatial-Logic-Aware Weakly Supervised Learning for Flood Mapping on Earth Imagery. **AAAI 2024**.
- A Hierarchical Spatial Transformer for Large Numbers of Point Samples in Continuous Space. **NeurIPS 2023**.

Preprints

- HyperMCTS: Hypergraph-Augmented MCTS for Long-Horizon LLM Agents. **arXiv 2026**.
- EarthSpatialBench: Benchmarking Spatial Reasoning Capabilities of Multimodal LLMs on Earth Imagery. **arXiv 2026**.
- Search in Imagination, Act in Reality: World-Model Test-Time Scaling for Interactive Agents. **Manuscript**.

Recognition, Service & Skills

Awards: ACM SIGSPATIAL Best Paper (2023); Gartner Graduate Fellowship (2023, 2025, 2026); AAAI Scholarship (2025); SDM Doctoral Forum Award (2024).
Reviewer: ICLR, ICML, NeurIPS, AAAI; IEEE TPAMI, TKDE, TNNLS.
Skills: Python, C++, PyTorch, Hugging Face, vLLM; supervised fine-tuning, distillation, synthetic data generation, agent planning and evaluation.